

LM1 Multiple Regression

多元参数解释

单元回归 $\begin{cases} \hat{Y} = \hat{b}_0 + \hat{b}_1 \cdot X \\ Y = b_0 + b_1 \cdot X + \varepsilon \end{cases}$

多元回归 $\begin{cases} \hat{Y} = \hat{b}_0 + \hat{b}_1 \cdot \hat{X}_1 + \hat{b}_2 \cdot \hat{X}_2 + \dots + \hat{b}_k \cdot \hat{X}_k \\ Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + \dots + b_k \cdot X_k + \varepsilon \end{cases}$

k = number of independent variables
 n = number of observations, n must $> k$

b_0 = intercept when all $X = 0$.
 b_1 = partial slope coefficient 偏斜率

★ 多元回归五大假设

- ε {
- 1 Linearity 线性
 - 2 Homoskedasticity 同方差 : $Var(\varepsilon) = \frac{\sum(\varepsilon_i - \bar{\varepsilon})^2}{df} = \frac{\sum \varepsilon_i^2}{df} = \text{常数, 平均偏离}$
 若违反, 则为异方差  ①
 - 3 Independence of errors 误差项独立性, $Cov(\varepsilon_i, \varepsilon_j) = 0$, 若违反, 则为序列相关 ②
 - 4 Normality 误差项服从正态分布 ($n \uparrow$)
- + 补充: 均值为0, $E(\varepsilon) = 0$. 即 $\sum \varepsilon = \sum \text{正} \varepsilon + \sum \text{负} \varepsilon = 0$.
 + $Cov(X, \varepsilon) = 0$
- ⑤ X 独立

误差项是
互不相关
正态分布
均值为0

3种违反情景

Heteroskedasticity
 Serial correlation
 Multicollinearity

多元回归特有

① X not random

② X 之间无完全(强烈)线性关系, 若违反, 则为 multicollinearity 多重共线性 ③

散点图识别 diagnostic 直观检验是否违反假设

- 1 { Scatterplots 研究 X, Y 之间, X, X 之间的关系
 Residual plots 研究 ε 相关 { 异方差 $\nwarrow$
 序列相关 $\nearrow$ } 注意 X, Y 轴

考点: 判断图片违反什么假设

② Scatterplots Matrix 多张散点图, 研究异常值, 多重共线性...

③ Q-Q plots 研究正态分布, 需拟合成直线 (S形 $\rightarrow$ 肥尾).



回归流程 详细检验

- 三大步骤
- ① 识别自变量、因变量, 找到合适模型.
 - ② 检验模型假设是否被违反, 修正
 - ③ 整体拟合度 goodness of fit $\rightarrow$ fit significant $\rightarrow$ best model?

Y是否连续 $\begin{cases} \text{Yes} \rightarrow \text{线性回归} \\ \text{No} \rightarrow \text{逻辑回归} \end{cases}$

Unstructured Analysis Step.

Text-based data

① Cleansing: HTML, Punctuations, Numbers, Spaces.

② Wrangling. Steps: ① Tokenization 断句.

② Normalization 大写→小写...变单词.

BOW: Bag of Words 单词打包.

N-grams 构成词组

③ DTM: Document Term Matrix 计算单词出现频率构建表格.

③ Exploration.

EDA 词云

Feature Selection 删掉用的高/低频.

Method: Frequency $\uparrow$ Chi-square t -值 $\uparrow$ Mutual Information $\uparrow$ 找重要.

Feature Engineering 贴标签. $\left\{ \begin{array}{l} \text{NER 专有名词} \\ \text{POS 词性} \end{array} \right.$

④ Model Training.

ROC. $\left\{ \begin{array}{l} \text{AUC} \uparrow \text{ better} \end{array} \right.$

样本内外越接近越好.

training data & cross validation data.

LM7 Big Data

3V: Volume / Variety / Velocity + Veracity 可靠性 / Value Source.

Structured Data Analysis Step: 首先 Unstructured Data 表格 → Structured Data.

- 1 Conceptualization of the modeling task
- 2 Data Collection
- 3 Data preparation and wrangling 清洗 & 预处理

- ① Cleaning
- Incompleteness error 不全
 - Invalidity error 范围不合理
 - Inaccuracy error 误差所问
 - Inconsistency error 内容不一致
 - Non-uniformity error 格式不一
 - Duplication error 重复

Handling outlier: OR

trimming 截尾
winsorization 截断
(用min or max 替代)

② Wrangling

Data transformations

- ✓ **Extraction:** A new variable can be created from the current variable for ease of analyzing and using for training the ML model.
- ✓ **Aggregation:** Two or more variables can be aggregated into one variable to consolidate similar variables.
- ✓ **Filtering:** The data rows that are not needed for the project must be identified and filtered.
- ✓ **Selection:** The data columns that are intuitively not needed for the project can be removed.
- ✓ **Conversion:** The variables can be of different types: nominal, ordinal, continuous, and categorical.

- ① Transformation 变形 5 方法
- ② Scaling 缩放 X 范围

Normalization/标准化
Standardization 正态分布

④ Data Exploration 人工简单分析

① EDA: Exploratory Data Analysis 传统方法描述数据

② Feature Selection 特征选择

③ Feature Engineering 特征工程 便于分析

影响 Model Performance

⑤ Model Training

① Method selection: 上-章

② Performance Evaluation.

a Error Analysis

Confusion matrix

Actual Training Labels			
		Class "1"	Class "0"
Predicted Results 机器学习预测	Class "1"	True Positives (TP) 拒伪	False Positives (FP) Type I Error 拒真
	Class "0"	False Negatives (FN) Type II Error 取伪	True Negatives (TN) 存真

• **Precision** is the ratio of correctly predicted positive classes to all predicted positive classes. 第一行

$$\text{Precision (P)} = \frac{TP}{TP + FP} \quad \text{阳性} \rightarrow P \downarrow$$

• **Recall (sensitivity)** is the ratio of correctly predicted positive classes to all actual positive classes. 第一列

$$\text{Recall (R)} = \frac{TP}{TP + FN} \quad \text{阳性} \rightarrow R \downarrow$$

• **Accuracy** is the percentage of correctly predicted classes out of total predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

• **F1 score** is the harmonic mean of precision and recall. 精确、召回和平均

$$\text{F1 score} = \frac{2 * P * R}{P + R} \quad \text{Harmonic mean}$$

b ROC Receiver Operating Characteristic

Area Under the Curve AUC 越大越好

c RMSE 样本外 E 标准差. $RMSE = \sqrt{\frac{\sum (E_i - \bar{E})^2}{n}}$ 越低越好

d Tuning 调优.

超参数 { 单个 confusion matrix
多个 Geilng Analysis

② Unsupervised 非监督式

unlabeled data

types

- 降维 Dimension Reduction: ↓k, 以防止 overfitting / 保留数据方差大, 多样性, 防止 underfitted
算法: PCA, 主成分分析法, k-L, composite variables 筛选
- 聚类 Clustering: 先总结, 再分类
算法:
 - K-means Clustering: 人为确定 hyperparameter k
 - Hierarchical Clustering: 无须确定 k

③ Neural networks 神经网络 (模拟人)

机制: Input / Hidden / Output Layer 人为设置节点数量

建模: Forward Propagation → 调试: Backward Propagation or Revision for hyperparameter

types

- Deep learning: hidden layer > 2
- Reinforcement learning: 从错误中总结经验

ML Algorithm Type

Variables	Supervised (Target Variable)	Unsupervised (No Target Variable)
Continuous	Regression - Linear; Penalized Regression/LASSO - Logistic - Classification and Regression Tree (CART) - Random Forest	Dimensionality Reduction - Principal Components Analysis (PCA) Clustering - K-Means k 值数 - Hierarchical 无k
Categorical	Classification - Logit - Support Vector Machine (SVM) 2 分法 - K-Nearest Neighbor (KNN) 多组 - Classification and Regression Tree (CART)	Dimensionality Reduction - Principal Components Analysis (PCA) Clustering - K-Means - Hierarchical
Continuous or Categorical	Neural Networks Deep Learning Reinforcement Learning	Neural Networks Deep Learning Reinforcement Learning

两类错误

Underfitting

Overfitting

无法 generalize well:

① 3 data Set to measure model

- ① Training Sample 训练集: used to develop model 建模 60% in sample 样本内
- ② Validation Sample 验证集: used to tuning model 调试 20% } out of sample 样本外
- ③ Test Sample 测试集: evaluating model 20%

② 3. errors

- ① Bias Error: underfitting → high in-sample error 欠拟合
- ② Variance Error: overfitting → high out-of-sample error
- ③ Base Error: $Y - \hat{Y}$, random noise / residual error

linear 易产生
需权衡
non linear 易产生

③ 解决 overfitting

- ① complexity reduction: penalty
- ② cross validation 滚雪球. (k-fold ⊗)

LM 6 ML 着重了解定性

Introduction

X (Features) Y (Target Variable) Hyperparameter (人为输入)

PET 定义法

- T: Task
- E: Experience / Error
- P: Performance Measure

无人工参与

ML 与 回归 相同点

- Find the pattern
- Apply

	ML	VS	回归
分门别类	x		✓
n	big data	Structured Unstructured	中/小
关系	linear/non-linear		linear
复杂度	多维		低维
X	features		自变量
Y	target variable		因变量
	hyperparameter		人为输入

Types of ML

算法
适用条件

① Supervised (SL) 监督式

labeled training data, 知道 Y 特征

types

- Regression model: continuous target variable
 - 线性 LASSO: 惩罚回归
 - 非线性 regularization → 控制 K, 防止 overfitting
- Classification model: Y only 分类, 定性不定量
 - 线性
 - 2类 SVM
 - 多类 KNN
 - 非线性: CART → Random Forest.

5 算法: ① Penalized Regression

线性: LASSO. OLS: SSE_{min} VS. LASSO: $[SSE + \text{penalty term}]_{min}$

非线性: Regularization 正则化: $K \downarrow$.

② SVM

a. 机制: n 维空间, n-1 hyperplane 超平面切分

- Support vector 离边界最近的值.
- margin 越大越好, 数据分割开
- discriminant boundary 分割线 (超平面).

b. soft margin (曲线).

c. 适用条件: ① linear data ② 二分类 ③ Small 且高维 data

③ KNN

a. 机制: ① Hyperparameter: $k=2$ ② 找数据 ③ 少数服从多数

b. 缺点: ① $k \leftarrow \begin{matrix} K \\ k \downarrow \end{matrix}$ 被平均 outlier 易受 K 影响.

② 相似定义认识不同.

④ CART

a. 机制: 决策树.

- Classification: 少数服从多数
- Regression: 取平均.

b. 优缺点: Visualize → 无 black box.

分支过多 overfitting 解决 regularization pruned 减枝

⑤ Ensemble

heterogeneous learners 不同算法

homogeneous learners 同质学习 bootstrapping

↳ Random Forest: 每棵树只包含部分 X, 不过于复杂. → 结果 wisdom of crowd. black box.

自相关 Autocorrelation 针对 ε

LM3 中学过, DW test / BG test 不适用

① 检验: ① 计算 ε 自相关系数

② AR 模型用 T-test 检验

② 后果: AR 模型特殊后果 — 季节性

③ 修正: 加入滞后项作自变量

adds seasonal-lag in AR model (AR(1) $\rightarrow$ AR(2)).1. 设假设 $H_0: \rho=0, H_a: \rho \neq 0$ 2. T.S. (t) = $\frac{r-0}{1/\sqrt{n}}$ 3. CV (t分布) $df=T-k-1$, 双尾.4. $|T.S.| > CV \rightarrow \times H_0 \rightarrow$ 自相关Standard error = $\frac{1}{\sqrt{n}}$ 条件异方差 ARCH 针对 $\varepsilon \rightarrow$ Variance of ε is not constant.① 后果: $PC(I), PC(II)$ 均有可能 $\uparrow$ ② 检验: ARCH(1): $G_t^2 = a_0 + a_1 \varepsilon_{t-1}^2$ $\varepsilon_{t-1}^2 \uparrow \rightarrow \hat{y}_{t+1} - y_{t+1}$ 偏离 $\rightarrow G_t^2 \uparrow$

T-test

VS.

volatility clustering

多元回归用 BP Test

①. $H_0: a_1=0$ $H_a: a_1 \neq 0$

② T.S. ③ CV ④ TS VS CV

⑤ $\times H_0 \rightarrow a_1 \neq 0 \rightarrow \varepsilon_{t+1}^2 \leftrightarrow \varepsilon_t^2$ $G_t^2 = \frac{\sum \varepsilon_t^2}{df} \rightarrow$ 异方差 a_1 significantly different from 0 $\rightarrow$ 条件异方差新方程: $\varepsilon_t^2 = a_0 + a_1 b_1 + a_2 x_t + u_t$ 1. 设假设 H_0 : 无异方差 H_1 : 有异方差2. Chi-square T: $\chi_{dp}^2 = N \times R^2_{\text{residual}}$ 3. Decision Rule: $R^2_{\text{residual}} \uparrow \rightarrow \times \leftrightarrow \varepsilon_t^2 \rightarrow \times H_0$ ③ 修正: Generalized least squares (GLS) $\uparrow$ 广义最小二乘法

评价 AR 模型

定量 $\leftarrow$ 样本内误差: $SEE = \sqrt{\frac{\sum \varepsilon_t^2}{n-k-1}}$, 越低越好样本外误差: RMSE (Root mean squared error) $RMSE = \sqrt{\frac{\sum (\varepsilon_i - \hat{\varepsilon}_i)^2}{n}}$, 越低越好定性 Instability $\leftarrow$ regime 区间 $\rightarrow$ 老数据, 新数据
sample period long, short

AR 以外, 其余模型

3 解即可 { ① Moving Average (MA) 移动平均模型 $\leftarrow$ 不断加入新数据, 赋予不同权重② ARMA 结合 AR 与 MA $ARMA(p, q)$ $\leftarrow$ p : autoregressive terms
 q : moving average terms

③ 2 组时间序列数据

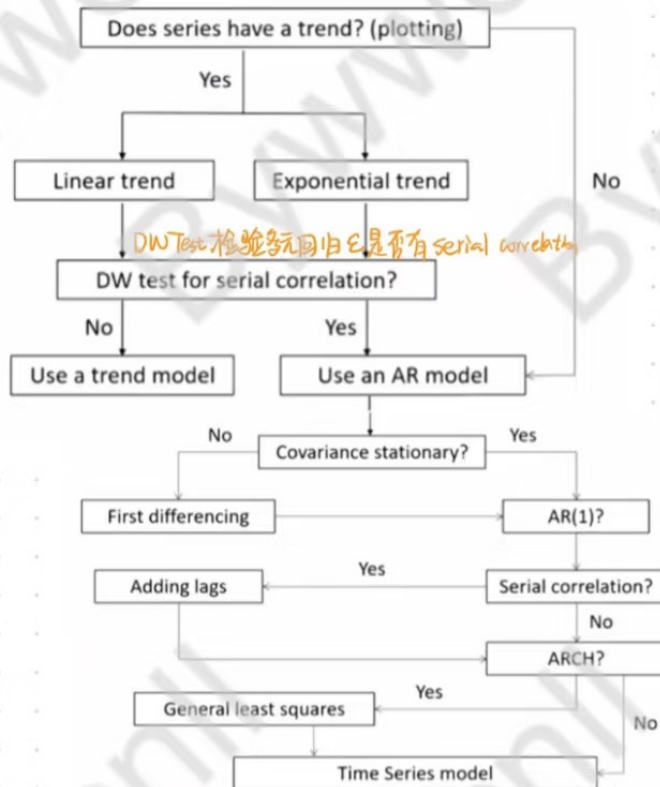
判断 unit root 时:

1 组: DF test

2 组: DF-EG test 协整即可 test cointegrated (H_0 : no coin..., H_a : coin...)

能否进行多元回归	Scenarios		能否
	None of the time series has a unit root		✓
	One time series has a unit root while another time series does not		✗
	Each time series has a unit root: whether the time series are cointegrated?	cointegrated	✓
		no cointegration	✗

LM 5 时间序列分析



Trend

- 分类
 - Linear Trend y_t 等差, 散点图成线.
 - Log-linear Trend y_t 等比, q 恒定 $\ln(y_t) = b_0 + b_1 \cdot t + \varepsilon_t$

- 缺点: 序列相关, 由 DW test 检验

- 修正: Autoregressive Models (AR) 自回归模型.

用过去自己解释今天自己

$x_{t-1} \rightarrow x_t$ AR(1) Model 滞后一期
 $x_{t-p} \rightarrow x_t$ AR(p) Model 滞后 p 期

- ① 计算: 递推法

- ② Assumptions
 - x_t covariance stationary 斜方差平稳.
 - errors are uncorrelated
 - homoskedasticity

斜方差平稳 针对 x_t

须满足

- 均值平稳 $\rightarrow$ mean reversion 均值回归.
- 方差平稳 Variance
- 斜方差平稳 Covariance

AR(1) 中: $x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$, $x_1 = \frac{b_0}{1-b_1}$ 长期稳定均值.
 if $x < \frac{b_0}{1-b_1}$, 上涨回归
 if $x > \frac{b_0}{1-b_1}$, 下降回归

若违反均值回归

- ① 后果: Random walk / unit root ($b_1 = 1$)
若 with a drift 则 + 漂移 undefined mean
- ② 检验: DF test
- ③ 修正: First-difference - 阶差分

$y_t = x_t - x_{t-1} = \varepsilon_t$ 验证新 y_t . 若仍不平稳. = 阶. = 阶...

1. 新方程 AR(1) $x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$ $x_t - x_{t-1} = b_0 + (b_1 - 1)x_{t-1} + \varepsilon_t$
2. 设假设 $H_0: g = 0$, unit root $H_a: g < 0$ 平稳且无 unit root 保证不发散
3. $T.S. = \frac{\hat{g} - 0}{s_{\hat{g}}}$
4. 查表 CV (修正)
5. $|T.S.| > CV \rightarrow \times H_0 \rightarrow b_1 \neq 1$ 长期稳定均值, 无 unit root

LM4 多元回归研究

Influence Analysis

① Influential Observation 强影响点 $\leftarrow$ X : high-leverage
可能导致 Y : outlier

② Detecting Influential Point:

单元: Scatterplot

多元: Quantitative way

老病在计算后判定

Measure	Y	X	Process	Is observation influential?
① Leverage		✓	h_{ii} ranges from 0 to 1	If $h_{ii} > 3(\frac{k+1}{n})$, then potentially influential
② Studentized residual 标准化残差	✓		Compare calculated t-statistic with critical t-value	If calculated t-statistic > critical t-value, then $df=n-k-2$ potentially influential
③ Cook's distance	✓	✓	Compare calculated Cook's D against $2 * \sqrt{k/n}$	If calculated Cook's D > $2 * \sqrt{k/n}$, then highly likely influential

$k+1$ 是 $\sum h_{ii}$

$\frac{k+1}{n}$ 是 h_{ii} 均值

$$t^* = \frac{e_i}{se} = e_i \times \frac{n-k-1}{\sqrt{SSE/(n-k-1)} - e_i^2}$$

删除一组观测值仅 $n-1$

$$|t^*| > CV(df=n-k-2) \text{ potential}$$

$$|t^*| > 3 \text{ outlier}$$

$$D_i = \frac{e_i^2}{k \times MSE} \left[\frac{h_{ii}}{(1-h_{ii})^2} \right]$$

$$D_i > 0.5 \text{ maybe}$$

$$D_i > 1 \text{ highly likely}$$

$$D_i > 2\sqrt{k/n} \text{ highly likely}$$

定性 X, Dummy Variable

n 种情况, $n-1$ 个 Dummy Variable. All Dummy Variable = 0, $\rightarrow$ Base Category.

Intercept Dummy
Slope Dummy 与 x 有联合作用. interaction term
Intercept + Slope Dummy.

定性 Y, 逻辑回归

Logit: odds for: $\frac{P}{1-P}$

考虑解读

log odds: $\ln(\frac{P}{1-P})$ 与 x 回归

ε 服从逻辑分布. 相比 Normal 肥尾 fat tail 现象

MLE 最大似然估计计算回归系数

由定性 Y 例推 P : 限 $Y=1$ 的概率

$$\ln(\frac{P}{1-P}) = b_0 + b_1 X_1 + b_2 X_2 + \varepsilon$$

$$\frac{P}{1-P} = e^{b_0 + b_1 X_1 + b_2 X_2}$$

$$P = \frac{1}{1 + e^{-(b_0 + b_1 X_1 + b_2 X_2)}}$$

检验 单个参数 t -test

整个模型 Likelihood Ratio (LR) test. 似然比检验, 类似于 Joint-F test. Test for fitness

Joint F $\varepsilon \sim \text{Normal}$

$H_0: b_1 = b_2 = 0$. H_a : 至少一个 $\neq 0$

$$T.S. = \frac{(SSE_{H_0} - SSE_{H_a})/2}{MSE}$$

LR $\varepsilon \sim \text{Logistic 分布}$

用 U 替代 SSE

1. $H_0: b_1 = b_2 = 0$. H_a : 至少一个 $\neq 0$

$$2. T.S. = -2 \times (U_{H_0} - U_{H_a}) \quad ||U|| \downarrow \rightarrow \text{好}$$

3. CV 卡方

4. $T.S. > CV$. xH_0

5. xH_a . 至少一个 $\neq 0$. 无限制 Model 好

U 为负, $||U|| \downarrow$ 好

LM3. 回归错误

建模错误 违反假设更严重, 影响 Consistency

Failures in Regression Functional Form	Explanation	Consequence
Omitted variables	One or more important variables are omitted from the regression. $E\epsilon_i \neq 0$	May lead to heteroskedasticity or serial correlation
Inappropriate form of variables	Ignoring a nonlinear relationship between the dependent and independent variable	May lead to heteroskedasticity 及 序列相关 Scatter plots 将丁变形
Inappropriate variable scaling	One or more regression variables may need to be transformed before estimating the regression	May lead to heteroskedasticity or Multicollinearity 转为 Common-Size
Inappropriate data pooling	Regression model pools data from different samples that should not be pooled	May lead to heteroskedasticity or serial correlation

好模型标准

- Principles for Proper Regression Model Specification
- Model should be grounded in economic reasoning.
 - Model should be parsimonious. 简洁
 - Model should perform well out of sample. generalised
 - Model functional form should be appropriate.
 - Model should satisfy regression assumptions.

1. 后果: $\begin{cases} 1. COV(X_i, Y_i) > 0 & \begin{cases} b_1 \text{ 准确} \\ b_0 \text{ 不准确} \end{cases} \\ 2. COV(X_i, Y_i) < 0 & \begin{cases} b_0 \text{ 不准确} \\ b_1 \text{ 不准确} \end{cases} \end{cases}$
 2. 检验: $E\epsilon_i = 0$ 包含 X_2
 residual plots.
 3. 修正: 加入 X_2

违反情景

1 Heteroskedasticity (Conditional / Unconditional)

① 后果: 不影响: Consistency / $\hat{b}_i$
 影响: $S.E. \rightarrow T\text{-test, F-test}$ 结果不可靠

② 检验: Residual scatter plots

BP test Chi-square Test.

③ 修正: Use White-corrected SE 有标准误差稳定.

以下 Conditional 的

T: $\begin{cases} SEE \downarrow \rightarrow SBI \downarrow \rightarrow |TS| \uparrow \rightarrow \text{拒绝 } PCI \uparrow \\ SEE \uparrow \rightarrow SBI \uparrow \rightarrow |TS| \downarrow \rightarrow \text{拒绝 } PCI \uparrow \end{cases}$
 F: $\begin{cases} MSE \downarrow \rightarrow |TS| \uparrow \rightarrow PCI \uparrow \\ MSE \uparrow \rightarrow |TS| \downarrow \rightarrow PCI \uparrow \end{cases}$

1. 假设: H_0 : 无异方差
 H_1 : 有异方差
 or $P\text{-value} < \alpha$
 2. Chi-square: $\chi^2 = n \times R^2_{\text{residual}}$ 判断过程.
 新方程: $\hat{\epsilon}_i = a_0 + a_1 b_i + a_2 x_i + u_i$
 3. Decision Rule: $R^2_{\text{residual}} \uparrow \rightarrow X \leftrightarrow \epsilon^2 \rightarrow X \leftrightarrow MSE \rightarrow X H_1$

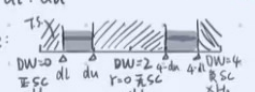
2 Serial Correlation / Autocorrelation ($\epsilon_i \> \epsilon_j$ 用 positive / negative) ($\epsilon_t \> \epsilon_{t-1} = \rho$ / $\epsilon_t \> \epsilon_{t-2} = \rho_2$)

① 后果

Independent Variable Is Lagged Value of Dependent Variable	Invalid Coefficient Estimates	Invalid Standard Error Estimates
① No	No	Yes 无须修正
② Yes	Yes	Yes

$\rightarrow SEE \downarrow \rightarrow |TS| \uparrow \rightarrow \text{拒绝 } PCI \uparrow$ 因为 ϵ 更平滑.

② 检验: DW Test

1. 假设: H_0 : No SC, H_a : VSC 3 解释即可!
 2. (7.5) $DW = \frac{\sum (\epsilon_i - \epsilon_{i-1})^2}{\sum \epsilon_i^2} = 2 \cdot (1 - \rho)$ DW 仅检验 ρ
 3. DW 表 cv. dl. du
 4. Decision Rule: $\begin{matrix} TS \\ DW \end{matrix}$ 
 $DW=0$ dl du $DW=2$ $4-dl$ $4-du$ $DW=4$
 $\epsilon_i = \rho \epsilon_{i-1}$ $\epsilon_i = \rho \epsilon_{i-2}$ $\epsilon_i = \rho \epsilon_{i-3}$

BG Test 用于检验多阶 SC.

1. 新方程: $\hat{u}_t = a_0 + a_1 x_t + a_2 x_{t-1} + p_1 u_{t-1} + e_t$ - 阶 $p=1$
 2. 假设: H_0 : $p=0$ H_a : $p \neq 0$
 3. 查表 近似 F 检验 ($p, n-p-k-1$)
 4. $TS > CU \rightarrow X H_0 \rightarrow SC$ ($u_t \rightarrow u_{t-1}$ 相关)

③ 修正: 修正标准误差

The corrections are known by various names, including serial-correlation consistent standard errors, serial correlation and heteroskedasticity adjusted standard errors (HAC), Newey-West standard errors, and robust standard errors.

3 Multicollinearity (多元回归特有).

① 后果: 不影响: consistency
 影响: S.E. 标准误差, 导致 T-test, F-test 结果不可靠 $\rightarrow MSE \uparrow \rightarrow SBI \uparrow \rightarrow PCI \uparrow$ 取伪
 $\hat{b}_0, \hat{b}_1$

② 检验

Classic: low T-test, high F-statistic (F 显著, 每个 T-test 不显著, 二者冲突) 可能存在

Pairwise Correlation $r = \frac{cov(X_i, X_j)}{S_x S_y}$ ($|r| > 0.7$) 可能存在

Variance inflation factor (VIF) 变量膨胀因子. $VIF = \frac{1}{1 - R^2}$

③ 修正

删自变量
 替代近似变量
 增加 n

$X_j = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \dots$ 回归
 $R^2 \uparrow \rightarrow R^2 \downarrow \rightarrow VIF \uparrow$

$VIF > 1$ 说明 $R^2 > 0$, no correlation
 $VIF > 5$ 须进一步研究
 $VIF > 10$ 严重共线, VIF 越大越严重

研究有没有线性关系

研究 2 个变量之间线性关系

研究一个变量跟多个变量线性关系

LM2 判定拟合度

指标 利用结果, 作出判断

1 ANOVA Table

	自由度	偏差度(平方和)	方差(均方)
	df	SS	MSS
Regression	k=1 x个数	RSS	MSR=RSS/k
Error	n-2 (n-k-1)	SSE	MSE=SSE/(n-k-1)
Total	n-1	SST=RSS+SSE	-

2 判定系数 R^2 (单元) $= \frac{RSS}{TSS} = 1 - \frac{SSE}{TSS}$ ($0 \leq R^2 \leq 1$)

R^2 (多元) $= 1 - \frac{SSE/(n-k-1)}{TSS/(n-1)} = 1 - \frac{\text{方差}(E)}{\text{方差}(Y)} = 1 - \left[\frac{n-1}{n-k-1} \cdot (1-R^2) \right]$ 可能 < 0.

① $\bar{R}^2 < R^2$ Adjusted for df. $\therefore k \uparrow \rightarrow R^2$ 不好. 造成过于复杂

② 新增 x. $|t\text{-statistic}| > 1$ $\bar{R}^2$ 可新加 x
 $|t\text{-statistic}| < 1$ $\bar{R}^2$ 不加 x.

3 AIC 赤池 & BIC 贝叶斯

① 好模型标准 ($\text{准确度} \uparrow \rightarrow \text{RSS} \downarrow \rightarrow k \uparrow$
 $\text{简洁度} \uparrow \rightarrow k \downarrow$)

② $AIC = n \ln \left(\frac{\text{Sum of squares error}}{n} \right) + 2(k+1)$
 $BIC = n \ln \left(\frac{\text{Sum of squares error}}{n} \right) + \ln(n)(k+1)$

③ BIC 惩罚力度更大, 更偏向简洁度高的模型 For Best Fit.
 AIC 更关注准确度. 用于预估. For Prediction Purpose

AIC & BIC 在二者间权衡

AIC & BIC 越低越好

假设检验

1 T-test 适用于单个参数检验 $b_i \neq 0, X_i \rightarrow Y$

2 F-test 适用于多个参数检验 $\left\{ \begin{array}{l} 9 \text{ 个 } x \text{ 斜率} \\ \text{Joint F-test, 判断新加入因素的解力度} \end{array} \right.$

3 个 Test 都是 $X \rightarrow Y$ 解释越好 or 显著! 整体模型 $\rightarrow$ General linear F-test

① $H_0: b_1 = b_2 = \dots = b_5 = 0$ 任何 $X \rightarrow Y$
 H_a : 至少一个斜率 $\neq 0, 1 \text{ 个 } X \rightarrow Y$ 显著 $\frac{(TSS-SSE)/k}{MSE}$

② $(T,S)F = \frac{\text{所有 } X \text{ 解释方差}}{X \text{ 解释方差}} = \frac{MSR}{MSE} = \frac{RSS/k}{SSE/(n-k-1)}$

③ 查表 cv. $\left\{ \begin{array}{l} F \text{ 分布 } (k, n-1-k) \\ \alpha = 5\% \\ \text{单尾 (右尾)} \end{array} \right.$ ④ $TS > CV, X H_0$ ⑤ 显著

① $H_0: b_4 = b_5 = 0$
 H_a : 至少一个斜率 $\neq 0$. $\frac{SS_{4,5} \text{ 平方和}}{df} = \frac{SSE_3 - SSE}{2}$

② $(T,S)F = \frac{X_4, X_5 \text{ 解释方差}}{X \text{ 解释方差}} = \frac{(SSE_3 - SSE_{4,5}) / 2 \text{ 个限制条件}}{MSE_{4,5}}$

③ 查表 cv. $\left\{ \begin{array}{l} F \text{ 分布 } (q, n-1-k) \\ \alpha = 5\% \\ \text{单尾 (右尾)} \end{array} \right.$ ④ $TS > CV, X H_0$ ⑤ 无限制解释力度更好

用模型预估

1 计算 $\hat{Y}$. 不用 $\pm E$

2 Confidence interval $\left\{ \begin{array}{l} \text{含 model error \& sampling error. 2种均无法消除} \\ \text{宽度} = (\hat{Y} + t_{sf}) - (\hat{Y} - t_{sf}) = 2t_{sf} \text{ 误差} \uparrow \rightarrow \text{准确度} \downarrow \rightarrow sf \uparrow \text{ 宽度越宽, 误差越大} \\ \text{单尾 (右尾)} \end{array} \right.$
 $2 \times t_{\text{信 standard error of forecast of } Y}$

1 Variation 偏差度 = SS 平方和.

2 Variance 方差 } = $\frac{SS}{df}$
 Mean Square 均方

3 SD 标准差 (Volatility) = $\sqrt{\text{方差}} = \sqrt{\frac{SS}{df}}$

标准差 SEE = $\sqrt{MSE} = \sqrt{\frac{SSE}{n-k-1}}$

$\uparrow \rightarrow$ 解力度 $\uparrow$